Matthew Murden

Computer Science Student

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Education

University of Birmingham, BSc in Computer Science with Year in Industry — Sept 2024 – June 2028

- Year 2: 65.7% (2:1)
- Key Modules: Software Engineering (78%), Functional Programming (78%), Security & Networks (73%)

Hymers College, A-Levels — June 2024

- Mathematics (A), Physics (A), Economics (B); Financial Studies AS-Level (A)

Experience

Intern Firmware Engineer, Honeywell Technologies – Industrial Placement — July 2026 – July 2027

Working on a next-generation, safety-critical ESSER fire alarm panel

- Engineered a Python tool to transfer files to embedded Linux CPUs over serial with base64 encoding and MD5 checks, cutting transfers from 30 minutes to 10 seconds.
- Diagnosed a failing internal tool and shipped a standalone fix, now used across a 15-engineer team.
- Driving the team's test automation with hardware-in-the-loop rigs (Arduino relays, photodiodes, machine vision, virtual HMI), turning weeks of manual testing into unattended overnight runs.

Web & Cyber Security Lead, TM Solicitors – Part-Time — June 2024 – present

- Led an authorised security audit of the firm's WordPress estate, uncovering two forgotten legacy installs and 39 CVSS-rated findings, including 9 vulnerabilities exploitable without login.
- Patched PHP2Shell (CVSS 9.2) within 2 hours of the fix's release and remediated a CVSS 9.8 plugin admin takeover the same day; both were actively exploited in the wild.
- Got a scam domain impersonating the firm taken down within 20 hours via multiple abuse channels, after the firm's own reports had failed, and hardened email and DNS security (SPF, DKIM, DMARC).

Founder & Director, Toned Designs — May 2025 – 2026

- Founded a UK streetwear brand with a registered trademark, generating £10k+ revenue from 250+ orders; built its custom Shopify theme (Liquid, JavaScript).

Projects

Seven-Pillar BTC Futures Trading System – Personal Project — Nov 2025 – Dec 2025

- Built an automated BTCUSDT perpetual futures system in Python where seven pillars (regime detection, LightGBM entry/exit models, risk sizing, safety layer, execution, monitoring) each own one decision, running live on the Binance demo account.
- Validated with walk-forward testing, fresh state per fold and a look-ahead leak test, then stress-tested realised trades against extra fees and slippage.

Polymarket Insider Trading Detection – Personal Research Project — Jan 2026 – present

- Built a live C++ pipeline streaming Polymarket data over WebSocket and matching it to on-chain settlement, measuring a ~2.2s lag between the two.
- Developed a Python anomaly scorer combining per-wallet signals into one risk score, validated against known insider cases, flagging clean control wallets at a lower rate than confirmed insiders.

Skills

Languages: Python, C++, Java, JavaScript/TypeScript, SQL

Systems & Tooling: Linux, Git, test automation, internal tooling, WebSocket data pipelines, embedded Linux, serial communication, hardware-in-the-loop testing

Security: Vulnerability assessment (CVSS), patch management, incident response, email and domain security (SPF, DKIM, DMARC)